

WHEN DOES TEST-TIME SELECTION HELP GENERATIVE ROBOT POLICIES?

DISENTANGLING CONSENSUS, FEEDBACK, AND RECOVERY

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ABSTRACT

Generative robot policies produce multiple plausible action chunks, but agreement among predictions need not identify a successful action. We examine test-time selection through complementary evaluations of action-space and decoder-space medoids on MIMIC-Video, GR00T N1.7, Xiaomi Robotics 0, and Cosmos Policy. We preserve the distinct benchmark populations and execution contracts rather than pooling their success rates. Three-seed GR00T results show mean decoder-medoid gains of 4.51 percentage points on SIMPLER and 3.65 on INT-ACT, but the corresponding Xiaomi effects are negative. In a matched Cosmos transfer study, decoder medoid and joint-value selection each achieve 112/180 successes, with seven rescues and seven harms. Decoder selection strongly favors particular fixed noise seeds, motivating controls that separate representation agreement from seed identity. Complementary feedback and physical-recovery studies distinguish selecting another prediction from obtaining another observation or changing the physical state. Large historical recovery gains are local: a corrected-runtime broad comparison finds no significant aggregate improvement. These results motivate a controlled evaluation of state risk, candidate quality, and intervention benefit as different quantities, not a claim of a universally better planner.

1 INTRODUCTION

Diffusion and flow-based robot policies generate temporally coherent actions by transforming stochastic noise into action chunks (Chi et al., 2023). World-action models additionally predict future observations, potentially providing a way to evaluate an action before execution (Kim et al., 2026; Pai et al., 2025). These interfaces invite test-time scaling: sample several continuations and choose a reliable one without changing the policy weights. However, similar samples can share the same mistake, an accurate failure monitor may not rank available actions, and requesting a new observation can change both computational cost and control behavior.

We study these distinctions through two complementary experimental programs. The first evaluates action-space and decoder-space medoids across multiple generative policies and distribution shifts. The second transfers decoder selection to Cosmos and compares changes to candidate selection, feedback frequency, and physical recovery. These programs share a scientific question and some algorithmic components, but not an interchangeable task population. Our comparisons therefore remain within matched model–benchmark contracts.

Self-consistency and representative-sample selection are established ideas (Wang et al., 2023; Dai et al., 2026; Rosasco et al., 2025). Our contribution is not the invention of medoids. Instead, we examine three practical questions: *Does agreement improve closed-loop success across seeds and models? Does a selector exploit state-dependent quality or fixed noise identity? How does selection differ from feedback and physical correction?* We report positive, null, and negative comparisons together with the controls needed to interpret them. This separates useful scoped findings from claims of broad robustness that the data do not support.

2 RELATED WORK

Generative policies and inference-time selection. Diffusion Policy models action trajectories directly (Chi et al., 2023); MIMIC-Video and Cosmos connect video representations to robot control (Pai et al., 2025; Kim et al., 2026). GR00T and Xiaomi Robotics 0 provide additional generative VLA interfaces (NVIDIA, 2026; Cai et al., 2026). KeyStone uses shared-context sampling and a guarded cluster-medoid selector (Dai et al., 2026); KDPE uses geometry-aware density estimation (Rosasco et al., 2025). Our global medoid, sample counts, action adapters, and seed schedules differ from those systems. Accordingly, our results are not a reproduction or refutation of their full published protocols. Small selector overhead also does not imply negligible multi-sample inference cost.

Prediction reliability and recovery. Ruan et al. connect action-state consistency to success and study future consensus, including deceptively consistent low-dynamics failures (Ruan et al., 2026). BadWAM demonstrates action–imagination decoupling under adversarial perturbations (Li et al., 2026). We examine natural and benchmark-induced failures, rather than adversarial attacks. Sentinel separates consistency from task progress (Agia et al., 2025); Rewind-IL combines failure detection with recovery (Zheng et al., 2026). These works motivate distinguishing the detection of difficulty from the benefit of a particular intervention. Our physical recovery consumes actions in the reached state; simulator branching is only an experimental measurement tool, not an online ability of the robot.

Evaluation under distribution shift. SIMPLER evaluates manipulation policies in simulation, INT-ACT probes generalization boundaries, and LIBERO and its extensions vary manipulation conditions (Li et al., 2024; Fang et al., 2025; Liu et al., 2023; Zhou et al., 2025; Fei et al., 2025). We keep standard LIBERO, LIBERO-PRO, LIBERO-Plus, SIMPLER, and INT-ACT separate. Confidence intervals and repeated-seed comparisons must also respect their experimental unit (Agarwal et al., 2021); an episode count is not the number of independent tasks or trained models.

3 PROBLEM AND METHODS

3.1 CANDIDATE GENERATION AND EXECUTION

At query q , let x_q denote the available real observations, robot state, and instruction. A frozen policy generates candidates

$$A_q^{(i)} = G_\theta(x_q, \xi_{q,i}), \quad A_q^{(i)} \in \mathbb{R}^{T \times d}, \quad i = 1, \dots, K. \quad (1)$$

The controller selects one candidate and executes its first $H \leq T$ actions. The metric may use a distinct scoring prefix $L \leq T$. Distinguishing T , L , and H prevents a change in execution horizon from being misattributed to a better selector.

For Cosmos, inputs are external and wrist RGB, nine-dimensional proprioception, and a precomputed instruction embedding. Each joint generation predicts a 16×7 action chunk, future observations/proprioception, and value. Our value baseline selects $i_V = \arg \max_i V_i$ from these joint samples. It is not the original paper’s rollout-finetuned autoregressive $a \rightarrow s' \rightarrow v$ planning procedure (Kim et al., 2026). After execution, ordinary queries receive actual images, not imagined ones.

3.2 ACTION-SPACE MEDOID

Write an action as translation u , orientation R , and gripper command g . After the model-specific action adapter, the control-aware distance is

$$d(a, b) = \frac{\|u_a - u_b\|_2}{\sqrt{3}} + \frac{0.5}{\pi} \theta(R_a R_b^\top) + 0.25 \mathbf{1}[\text{sign}(g_a) \neq \text{sign}(g_b)], \quad (2)$$

where $\theta(R) = \arccos(\text{clip}((\text{tr } R - 1)/2, -1, 1))$. For a model with 6D rotation output, R is obtained by orthonormalization; Cosmos uses native three-dimensional rotation-vector commands instead.

Translation is measured in the adapter’s command units: the $\sqrt{3}$ factor alone is not a physical normalization across embodiments. The temporally weighted distance and selected index are

$$D_L(i, j) = \frac{\sum_{t=1}^L 0.9^{t-1} d(a_{i,t}, a_{j,t})}{\sum_{t=1}^L 0.9^{t-1}}, \quad (3)$$

$$i_A = \arg \min_i \frac{1}{K-1} \sum_{j \neq i} D_L(i, j). \quad (4)$$

The chosen chunk is an actual sample, not the average of incompatible motions. This is a global medoid, distinct from KeyStone’s guarded dominant-cluster medoid. In the Cosmos transfer control, $L = 5$ while $H = 16$; this historical setting is explicitly retained rather than described as an executable-prefix match that was not used.

3.3 DECODER-SPACE MEDOID

The decoder variant compares aligned hidden action representations at the last captured conditional denoising call. For temporal tokens $z_{i,t}$,

$$D_Z(i, j) = \frac{\sum_t w_t [1 - \cos(z_{i,t}, z_{j,t})]}{\sum_t w_t}, \quad (5)$$

$$w_t = 4 \text{ for } t \leq 4, \quad w_t = 1 \text{ otherwise}, \quad (6)$$

$$i_Z = \arg \min_i \frac{1}{K-1} \sum_{j \neq i} D_Z(i, j). \quad (7)$$

Cosine norms are lower-bounded by 10^{-12} ; exact ties select the first index. GR00T excludes its state token and scores the generated action-token sequence, although only four actions are executed on SIMPLER. MIMIC extracts its action segment with its own decoder adapter. These implementations must not be interpreted as identical hidden tensors across architectures.

Cosmos requires a spatial adaptation. Its action latent frame has $16 \times 28 \times 28$ scalar elements containing 112 copies of a 16×7 action block. The last DiT hidden grid provides 196 spatial tokens of width 2048. A patch is not one action time. Let C_{pt} count decoded scalars belonging to time t within patch p . We use

$$\tilde{w}_p = \frac{\sum_t C_{pt} w_t}{\sum_{r,t} C_{rt} w_t}, \quad (8)$$

$$D_Z(i, j) = \sum_p \tilde{w}_p [1 - \cos(z_{i,p}, z_{j,p})]. \quad (9)$$

Features are captured read-only, without additional denoiser calls. The prefix-eight ablation sets weights of times nine through sixteen to zero.

3.4 SAMPLING CONTRACTS AND UNCERTAINTY

The decoder experiments use $K = 3$ and predefined seed groups. The inspected GR00T wrapper and Cosmos transfer reuse each candidate’s seed on every query, so observations change while noise identity is retained. This is a valid fixed-noise policy family, not fresh independent noise at every query. Our recovery experiments instead use $s_{q,i} = s_{\text{rollout}} + 1000q + i$ with $K = 4$. Comparisons across these contracts are not pooled.

Medoid centrality is neither variance nor a calibrated failure probability. For stochastic action vectors, a descriptive componentwise spread can be written as $U_A = (Hd)^{-1} \sum_{t,k} \text{Std}_i A_{i,t,k}$; the first-action version fixes $t = 1$. A scalar state-level spread does not rank candidates: subtracting the same $\lambda U(x_q)$ from every V_i leaves the value argmax unchanged. Candidate ranking requires a candidate-dependent score. Moreover, disagreement between distinct joint action–future samples mixes different plans; it is not repeated uncertainty about one fixed action’s consequence.

3.5 FEEDBACK AND PHYSICAL RECOVERY

Feedback replaces an unexecuted suffix using a fresh observation. Our scheduled physical recovery additionally localizes the target in external RGB using a separately trained DeepLabV3–ResNet50 heatmap localizer. The maximum heatmap location is projected through calibrated camera geometry and an object-conditioned height estimate. The gate checks finite values, heatmap score range, workspace/reach, and target–end-effector separation between 0.08 and 0.50 m. Score range is a heuristic, not calibrated confidence. The localizer uses 388 calibration images from 194 groups; consequently, the full system is not training-free even though Cosmos remains frozen.

An accepted correction opens and retreats, obtains another localization, approaches and descends, closes, and lifts. The stages allow at most $3 + 7 + 6 + 4 + 5 = 25$ physical actions, all charged to the 280-action episode budget. The policy then continues with $H = 8$. We compare this to H8 continuation, H16 continuation, and sensing/requery controls. Historical assessment times 56, 72, and 88 are scheduled ablations, not a universal learned failure time.

4 EXPERIMENTAL DESIGN

4.1 SEPARATE POPULATIONS, MATCHED COMPARISONS

SIMPLER-Bridge contains four tasks with 24 episodes each. The official INT-ACT Object-OOD subset contains eight tasks with 24 episodes each; the 16-task development extension is excluded. For GR00T and Xiaomi, the three reported primary seeds are 1, 2, and 3. Candidate groups are (1, 999, 998), (2, 997, 996), and (3, 995, 994). MIMIC uses the additional groups stated in Table 2; they are not interchangeable replicates.

The additional LIBERO experiments include standard Spatial with 100 episodes, GR00T’s PRO Spatial aggregate with 400 episodes across four OOD slices, and a first-100 Background Textures slice of LIBERO-Plus. These are not our Cosmos PRO Object-suite benchmark. The latter uses Object, Environment, and Position perturbations with 50, 50, and 99 valid cases, respectively. The decoder transfer grid instead contains 180 matched cases, 60 per factor, and three seed groups; its Position support is restricted to x0.2/y0.2. No headline SR is computed by merging these populations.

All Cosmos decoder comparisons generate 16 actions with five denoising evaluations per candidate and execute H16 unless labeled H8. A common initial state and cached first-query pool are shared between compared methods. Later observations and candidate pools naturally diverge after different actions. The 1440-rollout decoder program includes 720 main H16 rollouts, 360 fixed-candidate controls, and 360 H8 controls, not 1440 independent tasks.

4.2 OUTCOMES, UNCERTAINTY, AND AUDIT

The primary outcome is the simulator’s terminal task-success condition; timeouts are failures, whereas technical errors are missing data, not negative outcomes. We report counts and within-population SR. Micro-SR weights episodes equally; factor-macro SR weights perturbation factors equally. For paired outcomes,

$$\Delta\text{SR}_{\text{micro}} = \frac{N_{\text{rescue}} - N_{\text{harm}}}{N}. \quad (10)$$

Cross-model seed means are descriptive. Wilson intervals for individual binomial proportions do not replace a paired interval for a method’s effect. Exact McNemar tests are reported where paired outcomes were available; repeated observations of shared task/init groups require clustered analysis. Cosmos decoder intervals resample tasks within factors while retaining seed repeats. Broad recovery uses stratified task/init clusters, 5000 bootstrap replicates, 20000 sign flips, and Holm correction for four planned contrasts. An interval containing zero does not establish equivalence.

An audit identified missing solver warmstart and sensor-cache state in an earlier simulator snapshot implementation. Snapshot v2 stores complete MuJoCo integration state and observable caches. The broad v2 experiment includes 995 completed rollouts and explicit no-intervention replay checks. Historical scoped and shared-prefix studies are labeled separately; a broad v2 audit does not retroac-

Table 1: Complete three-seed comparisons. Entries are success counts for primary seeds 1/2/3; n is episodes per seed. The last columns are arithmetic mean SR across those seeds, not a pooled confidence interval.

Model	Suite	n	K1	Action	Decoder	K1 (%)	Decoder (%)
GR00T	SIMPLER	96	49/51/50	39/41/58	66/46/51	52.08	56.60
GR00T	INT-ACT	192	74/53/76	74/63/78	85/53/86	35.24	38.89
Xiaomi	SIMPLER	96	83/73/79	71/81/81	77/78/78	81.60	80.90
Xiaomi	INT-ACT	192	126/127/123	124/125/130	121/120/125	65.28	63.54

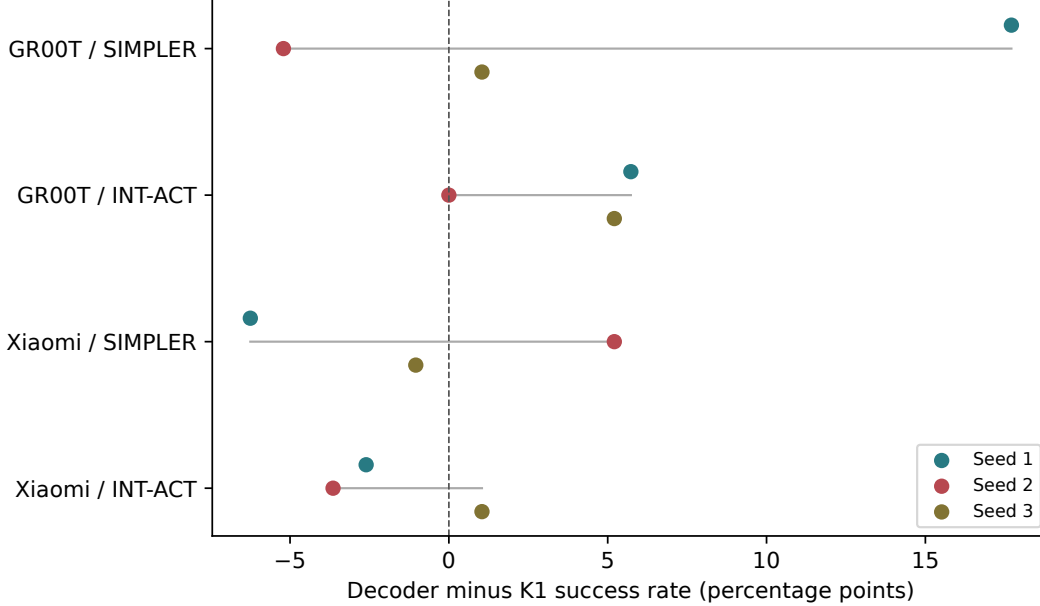


Figure 1: Decoder-medoid minus matched K1 success rate for every available seed in the complete three-seed groups. Dots are descriptive effects, not confidence intervals. The direction of benefit is not model-invariant.

tively validate those measurements. A new scoped snapshot-v2 replication is pending; it is not counted as a completed result in this manuscript.

5 RESULTS

5.1 CONSENSUS GAINS DEPEND ON MODEL AND SEED

Table 1 reports all three matched seed groups for the two models with complete three-seed SIMPLER and INT-ACT comparisons. Decoder medoid improves GR00T’s mean SR by 4.51 and 3.65 points, respectively. However, SIMPLER’s seed-specific effects are +17.71, −5.21, and +1.04 points. The first group’s 49/96 to 66/96 improvement has reported paired $p = 0.0076$, but is not a stable per-seed improvement. INT-ACT’s effects are more consistent in sign (+5.73, 0, +5.21 points), without an established aggregate significance claim from the available summaries.

Xiaomi shows no corresponding decoder improvement: its mean effects are −0.69 points on SIMPLER and −1.74 on INT-ACT. Action medoid is also heterogeneous, including a reported significant degradation for Xiaomi’s SIMPLER seed 1 ($p = 0.0169$). These observations caution against selecting the most favorable seed as a method’s representative result.

Table 2 summarizes the remaining slices. MIMIC’s INT-ACT baseline varies from 73/192 to 9/192 across seeds 0–2, making sampler and seed-matched controls essential. GR00T’s PRO decoder gain of 7/400 on seed 1 is accompanied by a loss of 3/400 on seed 2; the reported paired p values are

Table 2: Additional slices, kept separate. Counts separated by slashes in multi-seed rows are listed in the seed order specified. A dash is an absent matched result, not zero success. Different candidate groups are noted.

Model / slice	Sampling	K1	Action	Decoder
MIMIC / SIMPLER (96)	K1; action[0,1,2]; decoder[2,996,997]	45	52	42
MIMIC / INT-ACT (192)	primary0; candidates[0,1,2]	73	74	88
MIMIC / INT-ACT (192)	primary1 / primary2	33 / 9	–	–
MIMIC / INT-ACT (192)	action[2,996,997] / [3,998,999]	–	50 / 39	–
MIMIC / LIBERO Spatial (100)	reported default	69	75	64
GR00T / PRO Spatial (400)	primary1 / primary2 / primary3	246 / 244 / 246	248 / – / –	253 / 241 / –
GR00T / LIBERO Spatial (100)	decoder[2,997,996]	–	–	100
GR00T / Plus texture slice (100)	primary1; candidates[1,999,998]	100	100	100

0.0654 and 0.664. The Plus slice is saturated at 100/100 for every arm; this is neither a measured gain nor evidence about the complete benchmark.

5.2 DECODER AGREEMENT DOES NOT IMPROVE COSMOS RANKING

In the matched H16 Cosmos grid, first-candidate selection achieves 105/180, action medoid 110/180, and both decoder medoid and max-value 112/180. Decoder versus value yields seven rescues and seven harms, with task-cluster 95% interval $[-3.33, 3.33]$ points. Fixed second and third candidates achieve 110/180 and 109/180. The results do not establish extra utility from hidden-space selection over these seed controls.

The hidden selector nevertheless makes structured choices. For the three seed groups, a different favored seed is selected in 84.46%, 83.25%, and 81.18% of queries. Equal weighting of episodes preserves the pattern (85.90%, 84.99%, 82.70%), so it is not explained solely by longer failed episodes contributing more queries. This supports a hypothesis that hidden centrality partly reflects fixed noise identity. It does not prove that noise fully causes the observed performance or that every hidden selector behaves this way. Pooling candidate indices across groups would hide the preference.

5.3 MORE FEEDBACK IS NOT AN UNCONDITIONAL IMPROVEMENT

On the matched 120-case H8 subset, max-value H16 achieves macro-SR 65.00% versus 59.44% for H8. The difference interval includes zero. Full decoder H8 achieves 57.78%; prefix-eight decoder reaches 59.44%, matching its H8 value control rather than surpassing it. H8 roughly doubles logical candidate evaluations, with factor-wise ratios of 1.93–2.07. These are logical counts, not clean wall-clock latency measurements on the shared server.

A separate historical shared-prefix study on one Object task records 46/100 to 64/100 successes after a fresh-observation suffix replacement, with 31 rescues, 13 harms, and init-cluster interval $[4, 32]$ points. It uses 50 initial states and two repeats per state. This local finding and the broader H8 comparison concern different supports and are not contradictory. The historical replay contract requires a separate v2 audit.

5.4 PHYSICAL CORRECTION: STRONG SCOPE, LIMITED TRANSFER

On eight development-selected, calibration-familiar Position cells, historical recovery comparisons record 13/40 to 24/40, 6/40 to 25/40, and 16/64 to 41/64 successes relative to H8 continuation. The last effect is +39.06 points, with 27 rescues, two harms, and recorded init-cluster interval $[29.69, 48.44]$. A matched retreat/requery control achieves 8/40 versus full recovery’s 25/40. This supports a physical-correction mechanism within that population, subject to the scoped runtime check; it is not a latency-matched decomposition of every maneuver stage.

The transfer boundary is substantial: calibration-unseen x0.3 cells achieve 0/64 for continuation and 1/64 for recovery. In the corrected broad benchmark (Table 3), scheduled recovery gains only 0.68 macro points over H16, with interval $[-3.00, 4.36]$, nine rescues, and six harms. Its mean query count is 23.00 versus 13.86 for H16. The event-triggered arm performs no interventions in 199

Table 3: Corrected-runtime broad Cosmos comparison. Object/Environment/ Position have 50/50/99 cases. Macro-SR is the equal-factor average, not successes divided by 199.

Arm	Obj.	Env.	Pos.	Macro	/199
K1	96.00	40.00	29.29	55.10	97/199
Value H16	94.00	40.00	26.26	53.42	93/199
H8 at72	96.00	42.00	26.26	54.75	95/199
Recovery at72	90.00	42.00	30.30	54.10	96/199
Event	94.00	40.00	26.26	53.42	93/199

cases; its equal SR is a coverage failure of the tested trigger, not evidence for effective adaptive recovery.

6 DISCUSSION: WHAT SHOULD A RELIABILITY SCORE PREDICT?

For a fixed state s , candidate pool, horizon, and continuation policy, define $Q(s, A_i)$ as the success probability after executing candidate i and following that continuation. The selector’s shortfall decomposes as

$$1 - Q(s, A_i) = [1 - \max_i Q(s, A_i)] + [\max_i Q(s, A_i) - Q(s, A_i)]. \quad (11)$$

This identity separates limitations of the given pool/continuation from loss due to selecting its wrong member. It is not a new theorem or an already estimated oracle bound in every study. Reliable estimation requires repeated suffix outcomes and held-out evaluation of a chosen candidate; selecting and evaluating an oracle on the same outcomes is optimistic.

Three distinct targets follow: state risk $P(Y = 0 \mid s, \pi)$, candidate quality $Q(s, A_i)$, and recovery advantage

$$\Delta_R(s) = E[Y \mid \text{do}(R), s] - E[Y \mid \text{do}(C), s]. \quad (12)$$

A high failure-detection AUROC does not establish within-pool ranking or positive recovery advantage. Our geometric gate does not estimate Δ_R . Similarly, prediction error against an actual future observation is only available after execution and cannot be used as a pre-execution score for that same chunk. Predicted T16 futures must not be compared to H8 observations without temporal alignment.

The present evidence favors targeted controlled tests over additional coefficient searches on already inspected failures. Hidden-seed preference should be tested with fixed-candidate and fresh-noise controls; candidate quality should be measured within a common state/pool; recovery should be evaluated against its observation and action-budget controls. The positive findings in different cohorts do not yet prove that physical recovery rescues precisely the states in which consensus is uninformative.

7 LIMITATIONS AND CONCLUSION

The study combines complementary but heterogeneous model–benchmark experiments. Several cross-model results are supported by aggregate records rather than a common reprocessed episode-level archive. Three seed groups provide limited evidence about sampler generalization; some positive effects were identified during development. The P3 scope is calibration-dependent, its historical timing is scheduled, and its strongest historical measurements require corrected-runtime confirmation. We do not claim SOTA, universal consensus failure, a new backbone, or deployment on physical hardware.

Within these limits, the central finding is useful for designing inference-time methods: agreement, candidate quality, feedback value, and physical recovery are not interchangeable. Cross-model seed controls and matched interventions make their distinction observable. Reported local gains motivate further work, but neither a favorable seed nor an accurate monitor alone establishes a better closed-loop controller.

REPRODUCIBILITY AND AI USE

Tables and figures are generated from versioned aggregate or episode-level artifacts, with explicit support, seed groups, and runtime versions. The source package records evidence provenance and technical checks; it is not itself a release of all model weights or raw simulator artifacts. OpenAI ChatGPT/Codex assisted with literature synthesis, experimental tooling, analysis code, and drafting across the manuscript. Human authors are responsible for checking the scientific claims, references, and submitted artifacts. No generative model is listed as an author.

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